

❑ File & Directory Management

```
onworks@onworks: ~/student
onworks@onworks:~$ mkdir student
onworks@onworks:~$ cd student
onworks@onworks:~/student$ mkdir student1
onworks@onworks:~/student$ mkdir student2
onworks@onworks:~/student$ ls
student1 student2
onworks@onworks:~/student$
```

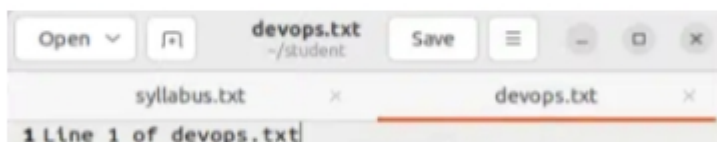
- **mkdir** → Creates a new directory (folder).
- **ls** → Lists files and directories in the current location

```
onworks@onworks:~/student$ rm syllabus
onworks@onworks:~/student$
```

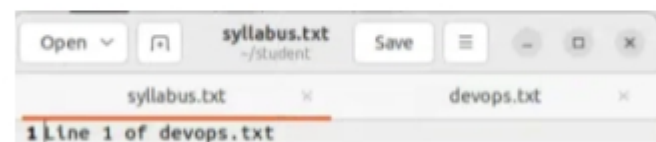
- **rm** → Deletes files or directories

```
onworks@onworks: ~/student
onworks@onworks:~/student$ ls
devops.txt student1 student2
onworks@onworks:~/student$ cp devops.txt syllabus.txt
onworks@onworks:~/student$ ls
devops.txt student1 student2 syllabus.txt
onworks@onworks:~/student$
```

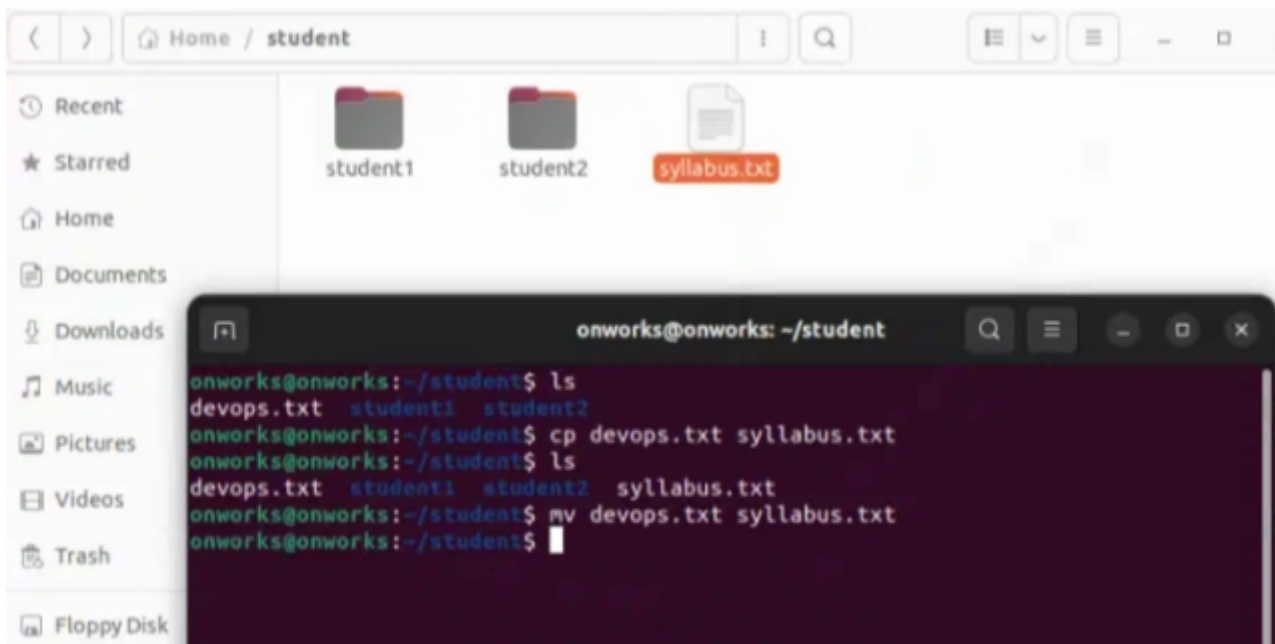
- **cp** → Copies files or directories to another location



^(copied text)

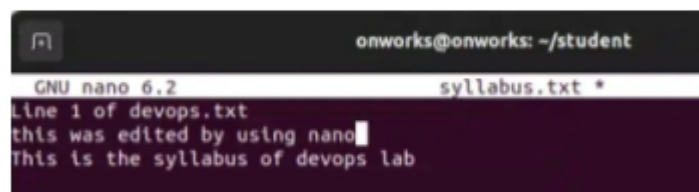
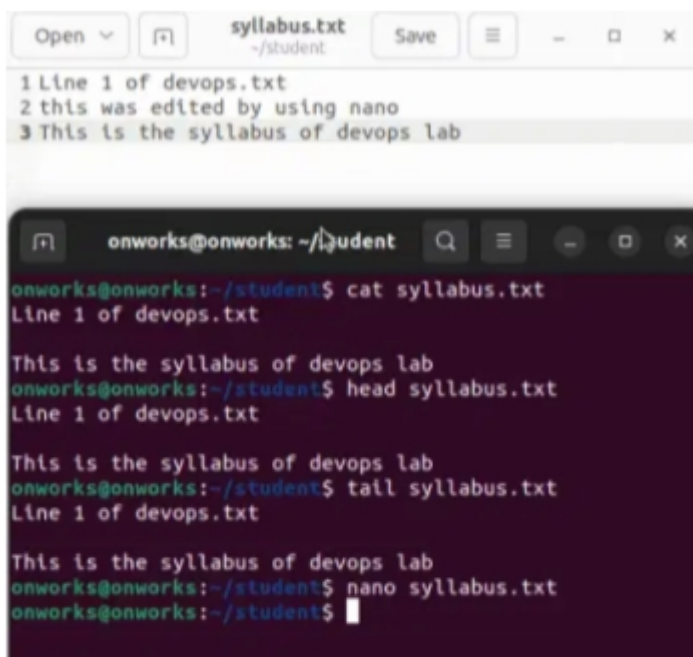


^(pasted text)



- **m v** → Moves or renames files/directories

□ Editing & Searching



- **nan o** → Opens a simple text editor in the terminal.

```
onworks@onworks: ~/student
onworks@onworks:~/student$ find /home -name syllabus.txt
/home/onworks/.local/share/Trash/files/syllabus.txt
/home/onworks/student/syllabus.txt
onworks@onworks:~/student$ grep 'This is the syllabus of devops lab' syllabus.txt
This is the syllabus of devops lab
onworks@onworks:~/student$ whoami
onworks
onworks@onworks:~/student$ uname -a
Linux onworks 6.2.0-26-generic #26-22.04.1-Ubuntu SMP PREEMPT_DYNAMIC Thu Jul 13 16:27:29 UTC 2 x86_64 x86_64 x86_64 GNU/Linux
onworks@onworks:~/student$
```

- **nan o** → Opens a simple text editor in the terminal. **fin d** → Searches for files and directories by name or pattern.
- **gre p** → Searches for text patterns inside files.
- **whoam i** → Displays the current logged-in username.
- **unam e** → Shows system information (OS, kernel version).

☐ Permissions & Processes

```
onworks@onworks: ~/student
onworks@onworks:~/student$ chmod 755 syllabus.txt
onworks@onworks:~/student$ ls -l
total 12
drwxrwxr-x 2 onworks onworks 4096 Jan 12 06:55 student1
drwxrwxr-x 2 onworks onworks 4096 Jan 12 06:55 student2
-rwxr-xr-x 1 onworks onworks  86 Jan 12 07:16 syllabus.txt
```

- **chmo d** → Changes file permissions (read/write/execute)
-

```
onworks@onworks:~/student$ ps -ef
UID          PID    PPID  C  STIME TTY          TIME CMD
root           1        0  0  06:42 ?        00:00:04 /sbin/init splash
root           2        0  0  06:42 ?        00:00:00 [kthreadd]
root           3        2  0  06:42 ?        00:00:00 [rcu_gp]
root           4        2  0  06:42 ?        00:00:00 [rcu_par_gp]
root           5        2  0  06:42 ?        00:00:00 [slub_flushwq]
root           6        2  0  06:42 ?        00:00:00 [netns]
root          10        2  0  06:42 ?        00:00:00 [mm_percpu_wq]
root          11        2  0  06:42 ?        00:00:00 [rcu_tasks_kthread]
root          12        2  0  06:42 ?        00:00:00 [rcu_tasks_rude_kthread]
root          13        2  0  06:42 ?        00:00:00 [rcu_tasks_trace_kthread]
root          14        2  0  06:42 ?        00:00:00 [ksoftirqd/0]
root          15        2  0  06:42 ?        00:00:00 [rcu_preempt]
root          16        2  0  06:42 ?        00:00:00 [migration/0]
root          17        2  0  06:42 ?        00:00:00 [idle_inject/0]
root          19        2  0  06:42 ?        00:00:00 [cpuhp/0]
root          20        2  0  06:42 ?        00:00:00 [cpuhp/1]
root          21        2  0  06:42 ?        00:00:00 [idle_inject/1]
root          22        2  0  06:42 ?        00:00:00 [migration/1]
root          23        2  0  06:42 ?        00:00:00 [ksoftirqd/1]
root          26        2  0  06:42 ?        00:00:00 [kdevtmpfs]
root          27        2  0  06:42 ?        00:00:00 [inet_frag_wq]
```

- **p s** → Displays running processes.

```
onworks@onworks:~/student$ top

top - 07:24:10 up 41 min, 2 users, load average: 0,46, 0,20, 0,18
Tasks: 186 total, 1 running, 185 sleeping, 0 stopped, 0 zombie
%Cpu(s): 30,6 us, 0,3 sy, 0,0 ni, 60,9 id, 0,0 wa, 0,0 hi, 0,0 si, 0,2 st
MiB Mem : 2892,3 total, 433,5 free, 827,7 used, 1631,1 buff/cache
MiB Swap: 3220,0 total, 3218,7 free, 1,3 used, 1845,8 avail Mem
```

PID	USER	PR	NI	VIRT	RES	SHR	S	%CPU	%MEM	TIME+	COMMAND
794	onworks	20	0	4303080	372332	136280	S	75,7	12,6	4:37.37	gnome-shell
10971	onworks	20	0	571348	53352	40720	S	2,0	1,8	0:03.50	gnome-terminal-
12114	root	20	0	0	0	0	I	0,3	0,0	0:00.08	kworker/0:0-events
1	root	20	0	167952	13132	8140	S	0,0	0,4	0:04.05	systemd
2	root	20	0	0	0	0	S	0,0	0,0	0:00.00	kthreadd
3	root	0	-20	0	0	0	I	0,0	0,0	0:00.00	rcu_gp
4	root	0	-20	0	0	0	I	0,0	0,0	0:00.00	rcu_par_gp
5	root	0	-20	0	0	0	I	0,0	0,0	0:00.00	slub_flushwq
6	root	0	-20	0	0	0	I	0,0	0,0	0:00.00	netns
10	root	0	-20	0	0	0	I	0,0	0,0	0:00.00	mm_percpu_wq
11	root	20	0	0	0	0	I	0,0	0,0	0:00.00	rcu_tasks_kthread
12	root	20	0	0	0	0	I	0,0	0,0	0:00.00	rcu_tasks_rude_kthread
13	root	20	0	0	0	0	I	0,0	0,0	0:00.00	rcu_tasks_trace_kthread
14	root	20	0	0	0	0	S	0,0	0,0	0:00.14	ksoftirqd/0
15	root	20	0	0	0	0	I	0,0	0,0	0:00.55	rcu_preempt
16	root	rt	0	0	0	0	S	0,0	0,0	0:00.01	migration/0
17	root	-51	0	0	0	0	S	0,0	0,0	0:00.00	idle_inject/0
19	root	20	0	0	0	0	S	0,0	0,0	0:00.00	cpuhp/0
20	root	20	0	0	0	0	S	0,0	0,0	0:00.00	cpuhp/1
21	root	-51	0	0	0	0	S	0,0	0,0	0:00.00	idle_inject/1
22	root	rt	0	0	0	0	S	0,0	0,0	0:00.20	migration/1
23	root	20	0	0	0	0	S	0,0	0,0	0:00.14	ksoftirqd/1

- **to p**→ Real-time view of processes and system resource usage.

```
onworks@onworks:~/student$ kill 1234
bash: kill: (1234) - No such process
```

- **kill**→ Terminates a process by its ID.

□ System Monitoring

```
onworks@onworks:~/student$ df -h
Filesystem      Size  Used Avail Use% Mounted on
tmpfs           290M  1,6M  288M   1% /run
/dev/sda3       29G   12G   16G  43% /
tmpfs           1,5G     0   1,5G   0% /dev/shm
tmpfs           5,0M   4,0K   5,0M   1% /run/lock
/dev/sda2       512M   6,1M  506M   2% /boot/efi
tmpfs           290M  184K  290M   1% /run/user/1000
```

- **d f**→ Shows disk space usage.

```
onworks@onworks:~/student$ free -m
              total        used        free      shared  buff/cache   available
Mem:           2892          828          432           44        1631        1845
Swap:          3219           1         3218
```

- **fre e**→ Displays memory (RAM) usage.

```
onworks@onworks:~/student$ whatis ping
ping (8) - send ICMP ECHO_REQUEST to network hosts
onworks@onworks:~/student$ date
Mo 12. Jan 07:35:39 CET 2026
```

- **whati s**→ Gives a one-line description of a command.

☐ Networking

```
onworks@onworks:~/student$ ping google.com -c 3
PING google.com (142.250.185.174) 56(84) bytes of data:
64 bytes from fra16s51-in-f14.1e100.net (142.250.185.174): icmp_seq=1 ttl=255 time=5.29 ms
64 bytes from fra16s51-in-f14.1e100.net (142.250.185.174): icmp_seq=2 ttl=255 time=4.91 ms
64 bytes from fra16s51-in-f14.1e100.net (142.250.185.174): icmp_seq=3 ttl=255 time=4.93 ms

--- google.com ping statistics ---
3 packets transmitted, 3 received, 0% packet loss, time 2002ms
rtt min/avg/max/mdev = 4.908/5.043/5.290/0.174 ms
onworks@onworks:~/student$ wget google.com/file.zip
--2026-01-12 07:27:02-- http://google.com/file.zip
Resolving google.com (google.com)... 142.250.185.174, 2a00:1450:4001:811::200e
Connecting to google.com (google.com)|142.250.185.174|:80...
```

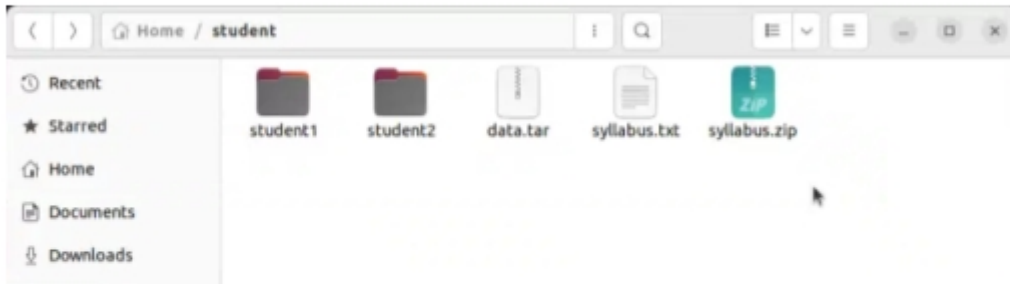
- **ping** → Tests network connectivity to a host.
- **wget** → Downloads files from the internet

☐ Package Management

```
onworks@onworks:~/student$ sudo apt update
[sudo] password for onworks:
sudo: a password is required
onworks@onworks:~/student$ sudo apt install git
[sudo] password for onworks:
sudo: a password is required
onworks@onworks:~/student$ sudo apt remove git
[sudo] password for onworks:
sudo: a password is required
onworks@onworks:~/student$
```

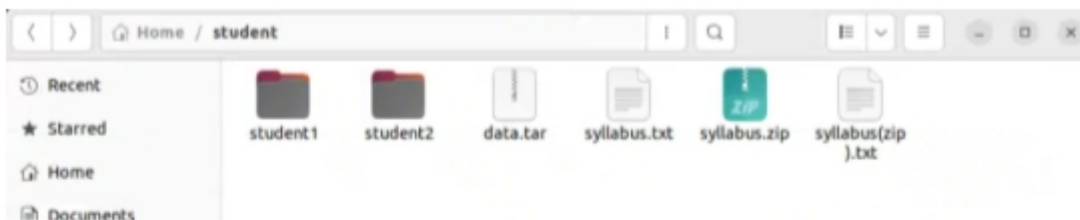
- **sudo** → Runs commands with superuser (admin) privileges.
- **apt update** → Updates package lists.
- **apt install** → Installs new software packages.
- **apt remove** → Uninstalls software packages.

□ Archiving & Compression



```
onworks@onworks:~/student$ tar -cvf data.tar syllabus.txt
syllabus.txt
onworks@onworks:~/student$ zip syllabus.zip syllabus.txt
  adding: syllabus.txt (deflated 13%)
onworks@onworks:~/student$ unzip syllabus.zip
Archive:  syllabus.zip
replace syllabus.txt? [y]es, [n]o, [A]ll, [N]one, [r]ename: r
new name: syllabus(zip).txt
  inflating: syllabus(zip).txt
onworks@onworks:~/student$
```

- **ta r** → Creates or extracts archive files.
- **zi p** → Compresses files into a **.zip** archive



^(unzipping the zip file {**syllabus(zip).txt**})

- **unzi p** → Extracts files from a **.zip** archive

❑ DevOps Tool

```
onworks@onworks:~/student$ git init
Command 'git' not found, but can be installed with:
sudo apt install git
onworks@onworks:~/student$ docker run -d -p 8080:80 nginx
Command 'docker' not found, but can be installed with:
sudo snap install docker          # version 20.10.24, or
sudo apt install podman-docker    # version 3.4.4+ds1-1ubuntu1.22.04.2
sudo apt install docker.io        # version 20.10.25-0ubuntu1~22.04.1
See 'snap info docker' for additional versions.
onworks@onworks:~/student$ sudo systemctl start jenkins
[sudo] password for onworks:
sudo: a password is required
onworks@onworks:~/student$
```

- **git** → Version control system for tracking code changes.
- **docker** → Platform for running applications in containers.
- **jenkins** → Automation server for CI/CD pipelines.

❑ User & System Commands

```
onworks@onworks:~/student$ w -h
onworks  tty2      tty2          22Aug23 49:25  0.01s  0.01s  /usr/libexec/gn
onworks  pts/1      -             22Aug23 873days 0.07s  0.03s  sudo su -
onworks@onworks:~/student$
```

- **w** → Shows logged-in users and their activities.

```
onworks@onworks:~/student$ exit --help
exit: exit [n]
Exit the shell.

Exits the shell with a status of N. If N is omitted, the exit status
is that of the last command executed.
onworks@onworks:~/student$
```

- **exit** → Closes the current shell session.

```
onworks@onworks:~/student$ shutdown --help
shutdown [OPTIONS...] [TIME] [WALL...]

Shut down the system.

Options:
  --help          Show this help
  -H --halt       Halt the machine
  -P --poweroff   Power-off the machine
  -r --reboot     Reboot the machine
  -h             Equivalent to --poweroff, overridden by --halt
  -k             Don't halt/power-off/reboot, just send warnings
  --no-wall       Don't send wall message before halt/power-off/reboot
  -c             Cancel a pending shutdown

See the shutdown(8) man page for details.
onworks@onworks:~/student$
```

- **shutdown** → Powers off or restarts the system.